



Solution Guide: Scheduling Backups, Cleanups, and Security Checks

Completing this activity required the following tasks:

- Moving the `backup.sh` and `update.sh` scripts to their corresponding system-wide `cron` directories. **Bonus** includes moving `cleanup.sh`.
- Creating `lynis` scripts to perform security scans.

Copy the scripts you wrote in the previous exercise to the appropriate `cron` directories in `/etc`. Specifically, your scripts should run at the following intervals:

- `backup.sh` should run weekly.
- `update.sh` should run weekly.

Navigate to your scripts folder and copy them to the corresponding `/etc` `cron` directory:

- `cd ~/Security_scripts`
- `sudo cp backup.sh /etc/cron.weekly`
- `sudo cp update.sh /etc/cron.weekly`

In addition to scheduling the preceding tasks, you should perform regular security scans to ensure your system hasn't been compromised.

- Create a script called `lynis.system.sh` in your `~/Security_scripts` directory. Write a command that will run a full-system scan using `lynis` every week that saves the results in `/tmp/lynis.system_scan.log`. Run:
 - `cd` to go to your home folder
 - `nano lynis.system.sh`

Unset

```
#!/bin/bash
```

```
lynis audit system >> /tmp/lynis.system_scan.log
```

- Create a script called `lynis.partial.sh`. Write a command that will use `lynis` to run daily scans for the test groups: `malware`, `authentication`, `networking`, `storage`, and `filesystems` that saves the results in `/tmp/lynis.partial_scan.log`. Run:

- `nano lynis.partial.sh`

Unset

```
#!/bin/bash
export TestGroup=('malware' 'authentication' 'networking'
'storage' 'filesystems');
for TG in ${TestGroup[@]};
do sudo lynis audit system --tests-from-group $TG >>
/tmp/lynis.partial_scan.log;
done
```

- Then, add both `lynis` scripts to the `root` crontab to create the tasks.

First:

- Ensure that the scripts are executable as follows:

Run:

- `chmod +x lynis.system.sh`
- `chmod +x lynis.partial.sh`

Run:

- `sudo crontab -e`, then add to the bottom:

Unset

```
@weekly lynis.system.sh
@daily lynis.partial.sh
```

- To use the `/etc/cron.<time>` route, run:
 - `sudo cp lynis.system.sh /etc/cron.weekly`
 - `sudo cp lynis.partial.sh /etc/cron.daily`

Bonus

1. Move the scripts you wrote in the previous exercise to the appropriate `cron` directories in `/etc`. Specifically, your scripts should run at the following intervals:

- `cleanup.sh` should run daily.

Navigate to your scripts folder and copy them to the corresponding `/etc` cron directory:

- `cd ~/Security_scripts`

- `sudo cp cleanup.sh /etc/cron.daily`

2. Explain why using scripts to run these commands is preferable to using a `crontab`.

- Multiple commands can be configured to run inside of a single executable script. This is convenient because there is no need to edit the `crontab` directly. You only need to edit the script directly, which will run during the next scheduled cronjob.
- Moving the `backup.sh`, `cleanup.sh`, and `update.sh` scripts to their corresponding system-wide `cron` directories.
- Creating `lynis` scripts to perform security scans.